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E-Mail Spam Classification via Machine Learning and Natural Language Processing

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***Abstract*—In our current modern scenario, majority of the correspondence and exchange in all business sectors take place through Emails. As the rate of exchange of information via emails is increasing exponentially, the amount of unsolicited bulk mail or Spam. These Emails are sent for a number of reasons: Extracting confidential information from individuals, promotion of adult content and marketing/advertising of products and services. Thus, keeping this mind, it is of paramount importance to build a comprehensive system for Spam Classification based on semantics based text classification using NLP and URL based filtering. Various Machine Learning algorithms have been surveyed and the objective is to create a model with high performance and efficiency.**

Keywords— Email Classification, Spam, Phishing, Machine Learning, Random Forest, Cyber Security, Text Classification, URL Classification, Add-on, Naive Bayes.

I. Introduction

In this era of globalisation, in the 21st century, majority of the correspondence and exchange in all business sectors take place via Emails. In the year 2019, [1] 246 billion emails were exchanged in a day and this figure is expected to grow to 320 billion emails by the year 2021. Out of these, 128.8 billion emails are business emails while 117.7 billion emails were consumer emails. Right from individuals to businesses and firms to even governments, every entity finds email communication to be a quite efficient, professional and effective way of transfer of information. It is highly possible that the content and body of these mails contain highly confidential information of high value to the entity. For a business, it could be the details of a prospective high profile deal which cannot be leaked to the general public, for an individual, it may contain the banking and account details of him or her, while for the government, information not in the nation's interest may get leaked. Thus, in this scenario it is of utmost importance that all the information exchanged is end to end encrypted There are 4 main types of spam emails which the paper has targeted namely: aggresive marketing, adult emails, phishing and email spoofing. The structure of the paper has been divided into four parts. First part consists of the literature survey, second part is the explanation of the various machine learning algorithms used, a comparative analysis of various algorithms, and the implementation of the machine learning model for text classification. In the third part, we performed URL Filtering, in the fourth part, we integrated both text classification and URL filtering to create an add-on on Gmail.

II. Literature Survey

In [2], an integrated approach involving all the three processes has more accuracy than any of the processes having a standalone approach (URL Analysis, NLP, ML). [3] further highlights the system of URL Classification and the Decision Tree algorithm is used, and the model is trained using a data-set from Phishtank. In [4], the URL is analysed based on parameters like the number of special characters and dots. Random tree and KNN have the same numerical value for the parameters but KNN requires more time to build the model than random tree, hence random tree is the most efficient machine learning algorithm used to classify the emails. In [5], Random forest, K nearest neighbour, decision making tree algorithm and support vector machines algorithms were implemented; Enron Spam Project data-set was used to train the model. Furthermore, an add-on to outlook was created in C in a visual studio. Random Forest was the machine learning algorithm which was the most accurate. Initially, the model is trained for text classification from the data-sets available on Enron Email Corpus and CMU Corpus. In [6] A total of 32 parameters are taken into consideration for file classification. Support Vector Machine algorithm is the most accurate for both the stages i.e. text classification and file classification. In [7], the drawbacks of models based on term frequency were considered which leads to huge computational load and slow training speed due to the size of huge feature vector space. Implementation of a model based on semantic similarity which extracts semantic meanings layer by layer using effective information retrieval techniques. In [8], performance of different ML algorithms by using features of keyword

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extraction, unigrams, bigrams and n-grams for classification of spam and fake online reviews. In [9],the measure of cosine similarity function was used and applied only on specific parts of speech (POS) and by employing lemmatization algorithms and effects of different pre-processing algorithms on classification accuracy. In [10], a semantic feature space was created from training data using statistical methods and solving the problems of conventional neural networks using BP algorithm. Keyword based spam filtering model efficient feature selection method using adaptive learning rate is used. In [11], modern day Spam Statistics and different types of Spam attacks (Email-phishing, spear phishing and spoofing) have been studies, existing softwares and scope of techniques for spam classification have been analysed, performance metrics of various Supervised Learning algorithms are compared and extraction of email routing information from a spam source. In [12], spam management for SMS services using text mining applications such as Rapid Miner for use in classification and clustering algorithms have been focused upon, they have justified the use of SVMs and Naïve Bayesian models as they have high performance even in the absence of large data and feature modelling and engineering. In [13], the main focus is on comparing four different machine learning algorithms for content based spam filtering technique, experimental results show that Neural Network classifier is more sensitive to the training set size and unsuitable for using alone as a spam rejection tool, generally, the performances of the SVM and RVM classifiers are less influenced by data sets and feature sizes, and obviously superior to the Naive Bayesian classifier. In [14] features for spam classification and spamming behavior of mails are discussed, the paper have presented a rule-based method for instantiating behavior-based features into discrete values, the paper presents the design and implementation of back-propagation neural networks for spam classification using behavior-based features. In [15] Spam classifier which is based on RF algorithm is used, they have used parameter optimization and feature selection , optimization of two parameters of Random Forest to maximize the Spam detection rates is done, and also provided the importance of individual feature selection. It can detect spam with low processing resources with high accuracy. In [16] research work on several State-of-the-art approaches used for various spam filtering methods. Different spam filtering formulas implemented by Gmail, Yahoo and Outlook; various performance evaluation measures are discussed on basis of which the performances are measured. All the machine learning algorithms used for filtering are discussed with their comparative analysis, the paper also includes various open research problems faced by existing techniques. [24] Highlights an algorithm where SVM and KNN both were implemented for Chinese web page classification, it improved the classifying predictability and gave a better feedback.

III. Methodology

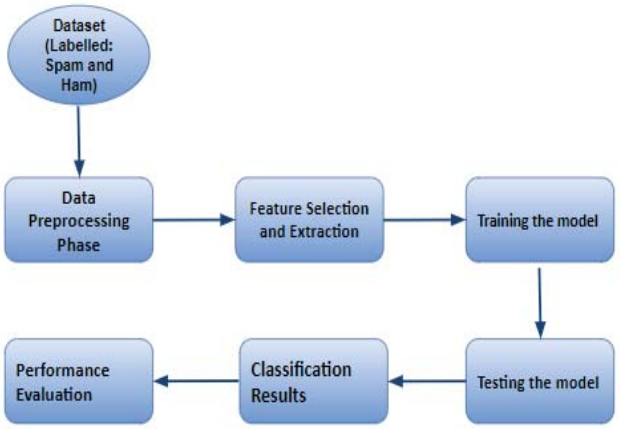


Figure 1: Functional Block Diagram

Figure 1 depicts the block diagram of the methodology and steps involved for text classification deploying the various machine learning algorithms

A. Machine Learning Algorithms

' There are various parameters involved in the machine learning algorithms which we have implemented which are as follows:

- Naive Bayes: Maximum likelihood estimate
- KNN: Euclidean distance
- Decision Tree: Number of nodes, class weight
- Random Forest: Maximum features
- Support Vector Machine: Decision function shape

1) Naïve Bayes: Naïve Bayes is a classification algorithm which uses probabilistic classifiers that are based on applying Bayes' theorem for conditional probability with two assumptions: all features are equally important and all features are independent of each other. For text classification, the features may be modelled on a multinomial distribution or if the features are continuous, they may follow a Gaussian distribution. A Naïve Bayes classifier deals efficiently with high dimensional data points [8] in the feature vector space and has faster training speed, it requires few training examples and it is widely used for text classification and with appropriate pre-processing of data its performance is comparable to that of advanced methods.

2) K-Nearest Neighbours (KNN): KNN is a widely used supervised learning algorithm mainly used for classification. It assumes that similar things are in close proximity to each other. The degree of similarity is calculated using similarity measures, usually the Euclidean Distance. KNN is easy to implement, since there is no need to build a model and tune parameters. KNN is a non-parametric algorithm so it makes no underlying assumptions about the distribution of the data. But as the dimensionality and size of the dataset increases the algorithm performs slower.

3) Decision Tree: Decision tree is a popular machine learning tool used for both classification and regression which has a flowchart like tree structure. It uses a recursive partitioning

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algorithm to test the presence of features or attributes based on certain purity indexes [8]. Gini Index and Entropy are the most popular indexes. Gini index is used to measure the probability that a randomly chosen feature would be incorrectly classified and Entropy is the amount of uncertainty which in turn is proportional to the information gain. These are used to decide which features are to be placed at the root or internal node. Decision tree works for both continuous and categorical variables. Figure 2 elucidates on how the decision tree may algorithm work, customised to our use case.

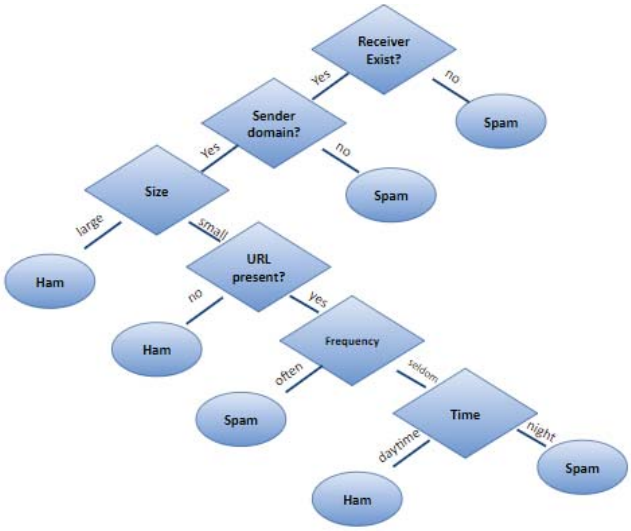


Figure 2: Decision Tree Flowchart

4) Random Forest: Random Forest is an ensemble learning algorithm which uses large number of uncorrelated decision trees to create a model which performs with a high accuracy on new datasets because of better generalization to new, unseen data. It reduces the variance and always prevents over-fitting of the model using the technique of Bootstrap Aggregation or Bagging where several subsets of the data are created from the training set and chosen randomly with replacement[19]. These subsets are then used to train each decision tree. Predictions by a multitude of uncorrelated models lead to a robust performance on new datasets. Figure 3 elaborates on the working of the random forest algorithm as a collection of decision tree

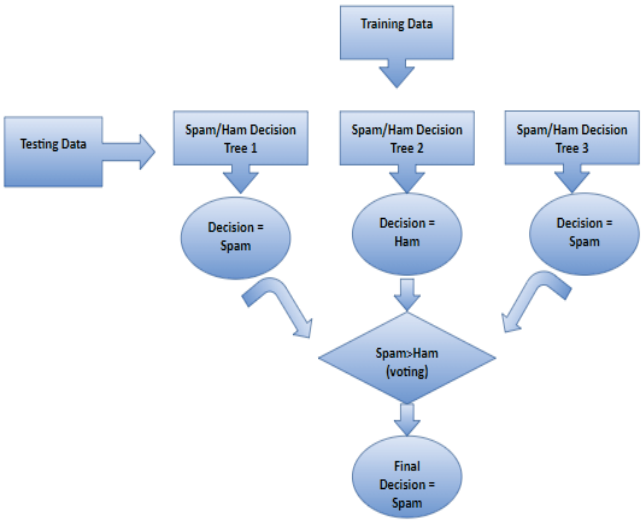


Figure 3: Random Forest Flowchart

5) Support Vector Machines (SVM): Support Vector Machine is a one of the most widely used supervised learning algorithm for classification. In a n-dimensional workspace a hyperplane based on the hypothesis of the model that segregates each of the classes from each other by analysing the features in the training data set [19]. SVM is a large margin classifier and it creates an optimum decision surface so that it has the maximum separation from the training sets of each of the classes. SVM has the unique ability to transform non-linearly separable data into linearly separable by the use of kernel methods. Kernel methods transform the training data such that the non-linear decision surface is transformed to a linear decision boundary in a higher dimensional space. Kernel methods enable to operate in higher dimensions without actually calculating the coordinates but rather computing the inner products between the vectors. This gives Support Vector Machines computational efficiency and the ability to scale to larger training sets.

B. Text classification

Text classification is also known as text-tagging and text-categorization, it is a process in which text which can be unstructured or structured is classified into organised groups and according to the requirements. Various machine learning models use Natural Language Processing to analyze text and perform other operation and then assign them group based on their content. Figure 4 depicts the text classification flowchart and the steps involved.

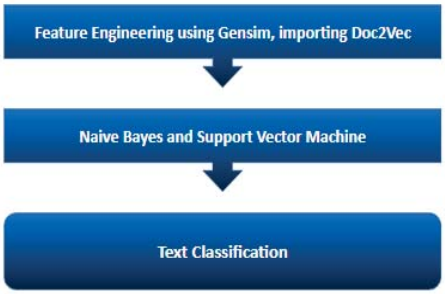


Figure 4: Text Classification flowchart

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- Feature Engineering
- Gensim Package
- Pre-processing the data
- Vectorization

1) Feature Engineering: Feature engineering has always been the key for creating superior, efficient and better machine learning models. It is very difficult to work on unstructured text data especially when trying to build a machine learning model. To convert the data into structured text data we need to process and transform noisy, unstructured data into a vectorized format, so that we can perform complex operations with the vectors and which can easily be understood by machine learning models. ML algorithms are based on three basic principles, statistics, optimization and mathematics. And therefore machine learning models are not intelligent enough to start processing text in unstructured and native form and that is why to obtain logical and understandable data, we perform feature engineering on the unstructured or raw data which helps us create feature vectors and these vectors are easily understood by machine as well as humans[19].

2) Gensim Package: Traditional methods for feature engineering were mainly based on counting the frequency of the word or the occurrence of the word and then predicting the output, example of such methods are term frequency(TF), term frequency-inverse document frequency (TF-IDF) etc. The above-mentioned methods are very effective but due to the nature of models, we are not able to consider important information like semantic similarity, the context of the data and the sequence around nearby words. So that is why we use more sophisticated models and for that, we require Gensim package. Gensim stands for Generate Similar, it is a popular open-source NLP library which is used for topic modelling. It is used to perform various complex tasks, for example, Semantic similarity approach, handling large text collections, and generation of word embeddings using Word2Vec to create feature vector space of 100 dimensions[19].

3) Pre-processing the data: There are several ways in which data can be cleaned and pre-processed in natural language processing pipelines. They are mentioned below:

- Removing tags: The text data set may contain unnecessary tags like HTML which are of no use and add zero value. So tags like these are removed.
- Removing accented characters: The text data set may contain accented characters, for example, é, á, ñ etc. And mainly in the English language data set. So over here these characters are converted into standard ASCII value, for example, 'á' is converted to 'a'.
- Expanding contractions: The text data set might contain shortened words or shortened syllables, and they are converted their expanded, original and standard form, example, 'I'd' is converted to 'I would', 'don't' is converted to 'do not' etc.
- Removing special characters: The text data set may contain special characters which are non-alphanumeric and are of no use, for example, $, @, etc. These are removed using regular expression(regex).
- Stemming and lemmatization: Word stems are the base form of the words that are created by adding suffixes or prefixes to the words, for example, dances, dancer, dancing all three have one root word which is dance. So the process of obtaining base word is known as stemming. Lemmatization is also similar to Stemming where affixes are removed to get root word and not root stem.
- Removing stopwords: There are many words in the data set which have little or no significance when creating features, such words have maximum frequency and their examples are a, an, the, is if etc. There is a stopword list available in a natural language processing tool kit.

There are various other pre-processing steps like tokenization, removing extra spaces, removing digits, text case lowering, spelling correctors etc[19].

4) Machine Learning algorithms: After comparing all the machine learning algorithms, two of them were used, and they are listed below:

- Naive Bayes:Naive Bayes is one of the ML model Or algorithm which is used, it is a statistical technique of email filtering and it uses BOW(bag of words) features which helps in identifying whether the mail is spam or ham. It works by correlating the tokens with spam and ham mails and then as mentioned in it's name it uses Bayes theorem and calculates whether the mail is spam or ham. It's main advantage is that it can be trained according to the users data set and it has very low false positive spam detection rate.
- Support Vector Machine: Support vector machine delivers very high accuracy result as it uses optimization process. It classifies by identifying the optimal hyperplane and maximizes the margin that separates spam and ham. SVM is robust in nature and is very effective when number of dimension is greater than number of samples[16].

5) Vectorization: To overcome the above-mentioned shortcomings we use VSM i.e. Vector space models, it is used in such a way that on the basis of contextual similarity and semantic similarity the vectors which are word vectors are plotted in the n dimensional vector space. It is known that words which occur and are used in the same context are semantically similar to one another. To achieve this we use word2vec models.

In simple words, word2vec is a technique for natural language processing, it uses a two-layered NN which is a neural network model,it then it learns word association from the large data set. And then it can predict words with similar meaning or can suggest words to complete the partial sentence. Word2vec model takes in a large number of text data and produces a vector space of hundred and more dimensions, and every word in the data set is assigned a unique corresponding vector and these vectors are positioned in a way that words with similar meaning or context are placed close to one another[19]. For the above process, word2vec uses 2 model architectures and

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they are,

- Continuous bag of words: Continuous bag of words is a model architecture that predicts the current word using the surrounding words or the words nearby.
- Continuous skip grams: Continuous skip-gram is a model architecture that predicts the surrounding words Or the nearby words of the sentence using the current word. Diagram [20]

C. URL Filtering

As aforementioned, the first step of spam classification is text classification in which the following machine learning algorithms i.e. Naive Bayes and Support Vector Machine having the greatest accuracy and precision are implemented in order to classify the text containing trigger words. In addition to this, the next step is URL classification and filtering. In order to protect the users from mails containing malicious URLs in the email body, it is of paramount importance that a proper methodology is followed to ensure data privacy, safety, integrity and security. This step has further been sub categorized into four steps as shown in the figure:

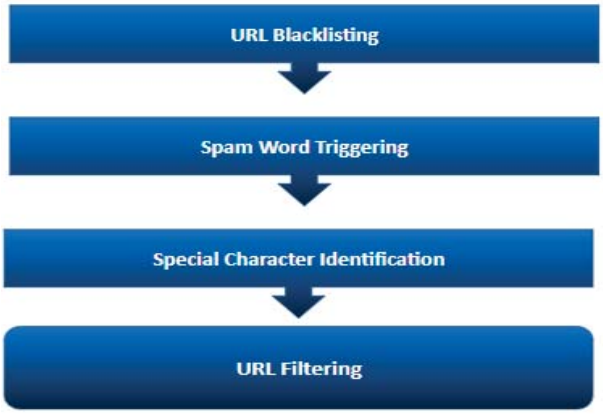


Figure 5: URL Filtering flowchart

- URL Un-Shortening
- Similarity to Blacklisted URL
- Presence of Trigger Words
- Identification of Special Characters

1) URL un-shortening: It may happen in some cases that a spammer may deliberately try to hide the actual harmful or malicious URL underneath a safe-looking URL in order to trick the user to dupe data or information. The hacker may perform this activity by shortening the original link to form a separate link which on the outside looks safe to visit. Thus this problem is tackled by un-shortening the URL in order to get the original link by allowing the user to get redirected unidirectionally to the original link. This link is then analysed in the further steps,

2) Blacklisted URL: The un-shortened URL along with the original shortened URL is compared to a list of URLs present in the blacklisted URL data-set, then the URL is automatically treated as malicious and the email is classified as spam.

3) Presence of trigger words: The URL is also analysed for some specific words known as the trigger words. Such words are mentioned in the spam trigger data-set [16] , if any phrase or word matches a word in the list, the URL will be classified as malicious, and the mail as spam. These words represent a variety of mails such as phishing, marketing and advertising, banking scams and adult words.

4) Identification of special characters: The last sub steps involve detecting special characters present in the URL. These characters are usually used by the attackers to form a URL similar to an already established safe looking URL so that the users can misinterpret this as a verified site and enter their personal confidential details, Such URL's mainly redirect the users to a website which has a similar User Interface to an already existing verified website in order to gain the confidence of the users, This phenomena is common in social networking sites or in the finance and banking sector. Thus, special characters like '@,' are identified in order to detect if a URL is malicious or not.

Thus in this step, if any of the four sub-steps has a positive output i.e. either of any aspect is present, the URL will be present in the mail and will be considered to be malicious.

D. Integration

After the model for the first step: text classification and the second step i.e. URL analysis and filtering were individually created, the next and the penultimate step was to integrate the two and create a model which takes into consideration the output of both,the first and the second step. An OR operation will be followed here. The model has been created such that if any of the two has an output saying that either the text has presence of spam words, or the URL is malicious, the mail will be considered as spam, and classified accordingly. Thus, for an email to be classified as ham, it will have to undergo a two step process, of text classification and URL classification. This model is then hosted as an API on the heroku platform, on the cloud server.

Google apps script is an application development platform used to create customized user interfaces that can integrate with the google workspace. Apps script is based on the javascript language and allows developers to create various types of add-ons.

Google has provided a platform for developers in order to test, deploy and publish add-ons and integrate them with other google products such as google docs, google sheets and Gmail.

For the purpose of a demonstration, a google workspace add on has been created which is integrated with Gmail. The apps script when executed classifies the mails in real time as spam or ham, by calling the API through a request.

IV. Experimental Results and Analysis

Initially, text classification was performed, as aforementioned using the Term Frequency Inverse Document Frequency approach. 5 different Machine Learning algorithms have been implemented in order to perform a comparative analysis of

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their performance taking into consideration the key performance indicators such as Accuracy, Precision, Recall and F1 score. A confusion matrix has been created to validate the same. Following are the algorithms implemented:

- K Nearest Neighbour
- Naive Bayes
- Decision Tree
- Random Forest
- Support Vector Machine

The algorithms are implemented on 2 different data-sets, namely spam.csv available on kaggle [23] and enron [22] spam data-set. Kaggle data set: The spam collection in this data set is a set of SMS tagged messages which we have used to perform experiment. This data set contains 5574 messages in english which are classified or tagged according to ham i.e. legitimate or spam[23]. Enron Data set: The Enron data set was used for performing experiment using different ML algorithms to get the required results. Enron data set in total consists of 30207 examples out of which we had 16545 examples classified as legitimate or ham and remaining 13662 examples were classified as spam[22].

A machine learning model was created and the following 5 algorithms were imported form the scikit learn library in and used for classification.

The comparative tabular analysis is given below as mentioned in the table:

Algorithm		Naive Bayes	Support Vector Machine	Decision Tree	Random Forest	K Nearest Neighbour			
	0/1					K=1	K=3	K=6	K=10
Precision	0	95%	98%	91%	96%	95%	93%	90%	89%
	1	97%	97%	83%	99%	99%	100%	100%	100%
Recall	0	100%	100%	99%	100%	100%	100%	100%	100%
	1	68%	86%	41%	74%	66%	50%	30%	24%
F1 Score	0	97%	99%	95%	98%	97%	96%	95%	94%
	1	80%	92%	55%	85%	79%	67%	46%	38
Accuracy	Model	95.48%	97.83%	90.90%	96.43%	95.29%	93.25%	90.58%	89.69

Table 1: Comparative Tabular Analysis 1

		Naive Bayes	Support Vector Machine	Decision Tree	Random Forest
Precision	0	94%	96%	96%	98%
	1	93%	73%	91%	97%
Recall	0	97%	91%	97%	99%
	1	85%	86%	89%	95%
F1 Score	0	96%	93%	96%	98%
	1	89%	79%	90%	96%
Accuracy	Model	93.65%	89.55%	94.70%	97.60%

Table 2: Comparative Tabular Analysis 2

		Naive Bayes (Multinomial)	Support Vector Machine	Naive Bayes (GNB)
Precision	0	92%	96%	90%
	1	96%	73%	92%
Recall	0	82%	91%	78%
	1	71%	86%	71%
F1 Score	0	84%	93%	76%
	1	82%	79%	84%
Accuracy	Model	95.8%	89.55%	89.8%

Table 3: Comparative Tabular Analysis 3

Furthermore, there are certain shortcomings in the TF IDF approach which are overcome by using the Gensim Library. In this paper, the TF-IDF approach was discarded while the gensim library was used which support semantic similarity yielding better results. Furthermore, in addition to text classification, the URL present was also classified via a three step filtering and analysis. This was the novelty presented in this paper. The Gensim library has been introduced by google in 2013 which utilizes the text8 data-set which has been iterated over 13 million documents, improving the accuracy. The gensim library takes into consideration the semantic similarity approach. As the accuracy for Naive Bayes and Support Vector Machine was highest, they were implemented along with gensim which resulted in a better accuracy as compared to TF-IDF approach. The accuracy for Naive Bayes and Support Vector Machine is 95.48 Percent and 97.83 Percent

In order to obtain real time results, 100 mails were sent to a particular Gmail account to check if the emails were classified correctly, the results obtained were as follows:

Total Number of Emails sent: 110
Spam Emails sent: 57
Ham Emails sent: 53
Spam Emails classified as spam (True Positive): 53

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Spam Emails classified as ham (False Negative): 4
Ham Emails classified as spam (False Positive): 3
Ham Emails classified as ham (True Negative): 50
Accuracy: 91.11 Percent
Precision: 94.64 Percent
Recall: 92.98 Percent
F1 score: 0.9137

V. Future Scope

Further research in this topic can be done across various sub-domains. Initially, the focus can be on improving accuracy by using some more computationally expensive but accurate machine learning classifiers like XGBoost. Furthermore, different word embedding algorithms other than Gensim word2Vec can be explored. Research in the field of deep learning could include transformer based deep learning models which was introduced in 2017. It enables training on humongous data sets, and also includes pre-trained systems which are used for text summarization and translation. Lastly, real time learning of email classifiers is something which the current data-sets do not focus on. It is important because real time factors play a huge role in determining the classification accuracy.

VI. Conclusion

A comprehensive and efficient spam classification system has been created which follows a two step methodology to completely ensure that the mail received is spam or not. Initially, text classification takes place which is followed by URL analysis and filtering in order to determine if any link present in the mail is malicious or not. For text classification, 5 machine learning algorithms were studied and analyzed, out of which Naive Bayes and Support Vector Machine having the highest accuracy were included in the final model. Various data-sets have been referred to for a list of spam trigger words and a list of blacklisted URLs. This model was hosted as an API which was then called by the javascript code in the google apps script in order to classify mails in real time in gmail.

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